

Fig. 3: Classification of 3D methods

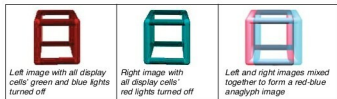


Fig. 4: A basic anaglyph. Consider the RGB of two filters
(Courtesy: www.google.com/image search)

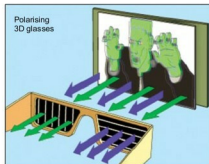


Fig. 5: Simple polarisation process
(Courtesy: www.google.com/image search)

or without glasses. Stereoscopic methods use glasses for producing the cognitive sensation. These methods further use time-parallel or time-multiplexing approaches.

In time-parallel approach, the left and right views are displayed simultaneously. For this, the viewers must wear wavelength-selective filters or polarised glasses.

In time-multiplexing approach, there is an alternate display of left and right views. Both the views are stored in two separate frame buffers and the screen is refreshed alternately. For time-multiplexing, liquid crystal shutters are used.

Autostereoscopic method. With

this method, one can project 3D content without glasses. This method comprises different displays that work on different principles such as integral imaging, holographic approach and use of voxels to project depth.

Stereoscopic method. The basic idea behind this is parallax. More the parallax, the more it creates stereoscopic sensation. Our binocular

vision enables us to have horizontal parallax, that is, the ability to view an object in two different ways horizontally.

The inter-ocular distance between eyes is about 6.35cm (2.5-inch), which forces the eyes to form slightly different images of an object, causing retinal disparity. The human brain then combines these 2D images into one perspective image called stereoscopic image.

Wavelength-selective glasses – Anaglyphs. In 1858, Joseph D'Almeida projected a 3D magic lantern by using red and blue filter glasses. However, Louis Du Hauron created the first printed anaglyph using colour printing available at that time. An anaglyph is a type of 3D image created from two photographs taken approximately 6.35cm (2.5-inch) apart.

Disadvantages.

1. High crosstalk: Colour filters are not perfectly matched
2. Anaglyphs lead to reduced perception of colours
3. Different people have different chromatic adaptation, which leads to eye irritation

Polarising glasses. The idea is based on different polarisations of the two

images. In this method, to overcome the reduced perception in anaglyphs, a projector projects two views onto the screen (having different polarisations).

But, for polarisation, we need to wear special glasses that would allow only one image into each eye. As the glasses contain polarised lenses, they would allow a specific orientation of light to pass.

However, the problem with this was that the polarised light from each row would create 1D for each eye, so the resolution would be half. The solution that came up was spatial multiplexing in which a checkerboard pattern in each pixel was used. The left pattern was opposite of right. This provided better sampling and thus better resolution than its counterpart.

LCD shutter glasses. This concept is based on real-time processing by the computer that provides control signals to the glasses.

In this case, the LCD and filters cause the lenses to lighten and darken at the refresh rate of computer. An IR sensor is used as an indicator for the glasses to judge the darkness and lightness of eye filters. This provides a much realistic image. The refresh rate is responsible for quality 3D.

For watching high-speed movies or sports in 3D, a refresh rate of about 200Hz is needed. Plasma TVs provide a very reliable option as they have a refresh rate of 600Hz. Many television and computer manufacturers are using this technique. The disadvantage is that all this depends on the refresh rate. Having a non-uniform refresh rate causes the screen to flicker, and thus very few monitors provide the required compatibility.

Autostereoscopic methods

The concept is based on glassesless 3D. For this, there are two possible approaches:

1. Eye tracking. For creating 3D as per eye position, the camera constantly observes head position, so the screen can be mechanically shifted. One approach, for example, is two-